ROC Curves & Confusion Matrix

Based on: "Model Assessment with ROC Curves" by Lutz Hamel,
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Task

Classification model assessment

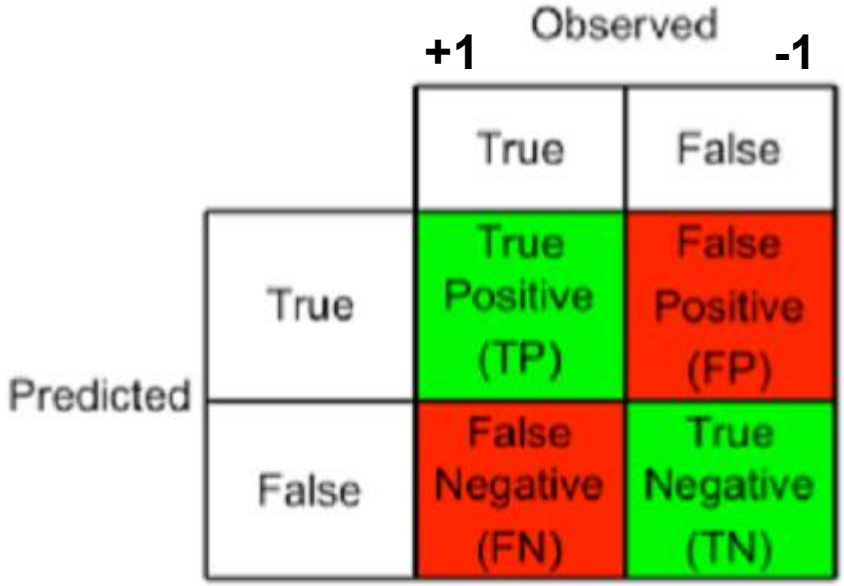
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Binary Classification: +1/-1

- Four possible situations with respect to observed-known and predicted:
  1. **T**rue **P**ositive. (**hit**)
  2. **T**rue **N**egative. (**correct reject**)
  3. **F**alse **P**ositive. (**false alarm**)
  4. **F**alse **N**egative. (**miss**)

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Confusion Matrix



Correct classifications

Errors

Note:

(TP + TN) + (FP + FN) = Correct classifications + Errors = # instances.

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Measures Based on Confusion Matrix

Accuracy:

Ratio of instances correctly classified.

$$\text{accuracy} = \frac{\text{correct classifications}}{\#\text{instances}} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}}$$

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Measures Based on Confusion Matrix

Precision:

Accuracy with respect to instances predicted as positives.

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

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Measures Based on Confusion Matrix

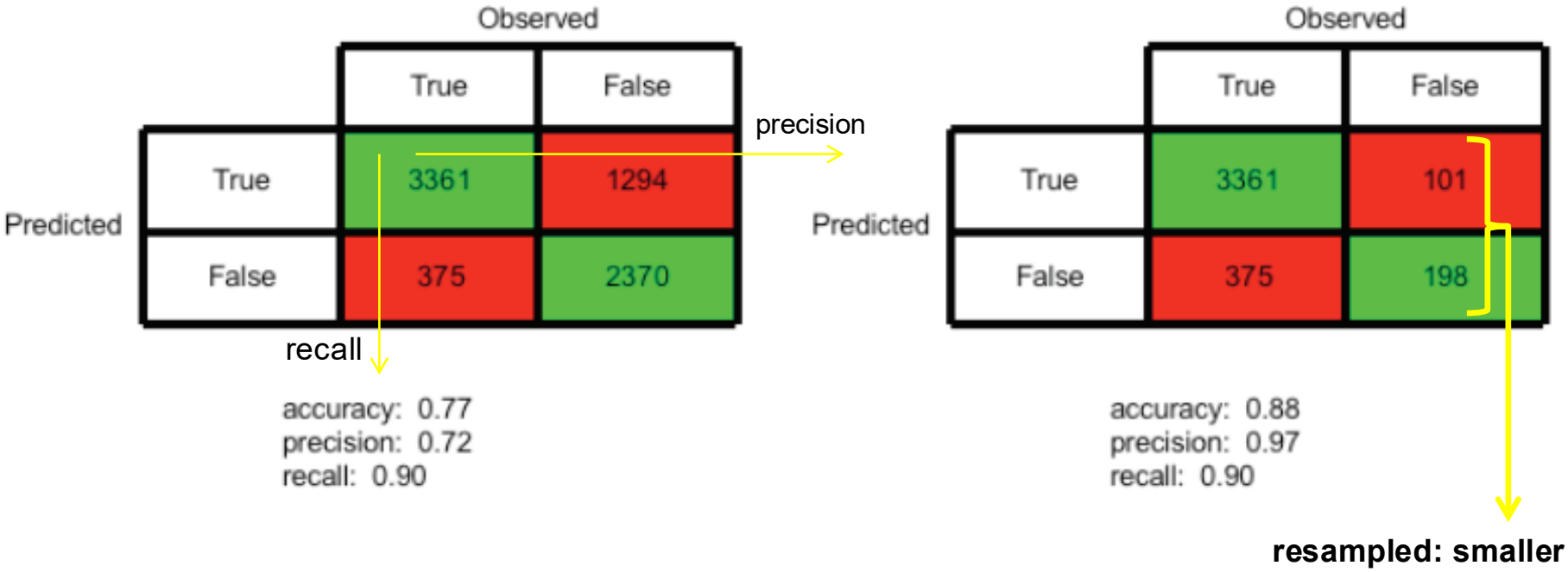
Recall:

Ratio of positive instances that are correctly predicted.

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

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Disadvantages



Accuracy-precision-recall:

- Depend on the choice of test set.
- Sensitive to class skew.
- They only produce a scalar as an assessment of the model.

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ROC Curves

- They are independent of the test data, at least with respect to class skew.
- Requires instance prediction-confidence for its construction:
  - Naïve Bayes, ANN, SVM and many others, establish some ranking about assigning instance to each class, then form a decision-prediction.
- Each ROC point is given by

  (false positive rate, true positive rate)

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ROC Curves: FPR/TPR

- False positive rate:

  (how many negatives perceived positive; minimize)

$$\text{false positive rate} = \frac{\text{FP}}{\text{TN}+\text{FP}}$$

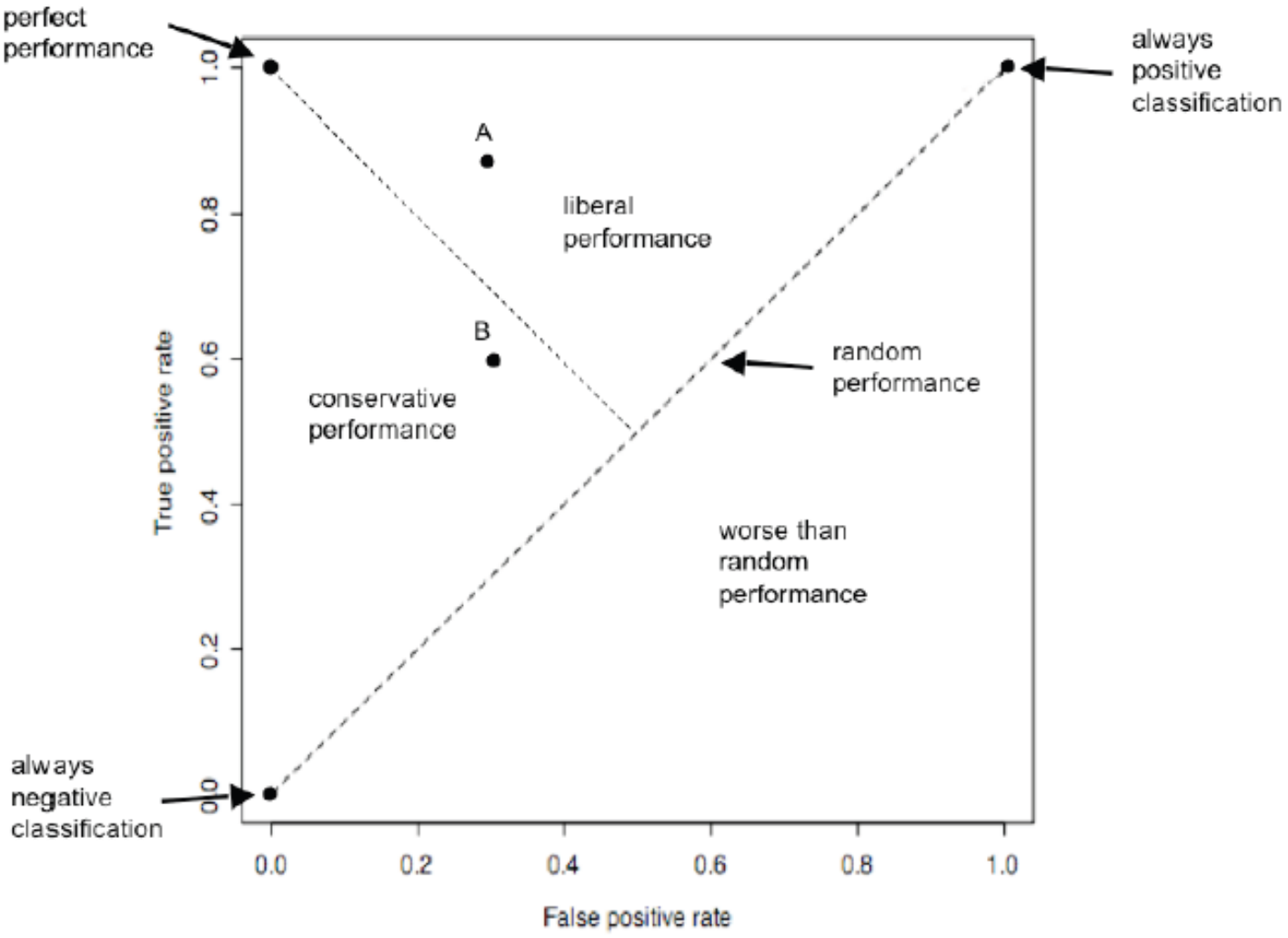
- True positive rate (recall):

  (how many positives perceived positive; maximize)

$$\text{true positive rate} = \frac{\text{TP}}{\text{TP}+\text{FN}}$$

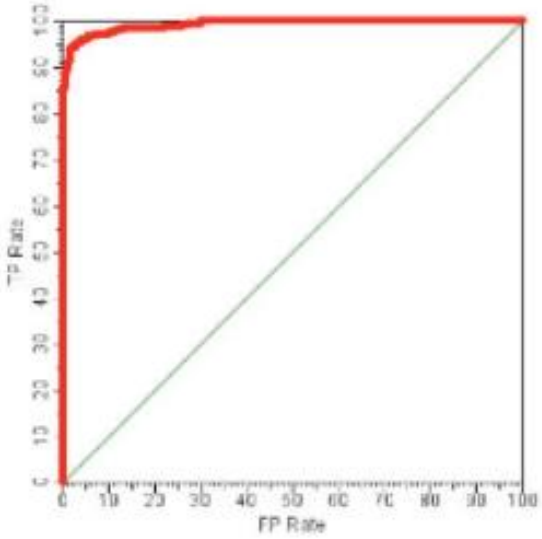
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ROC Graph: Regions

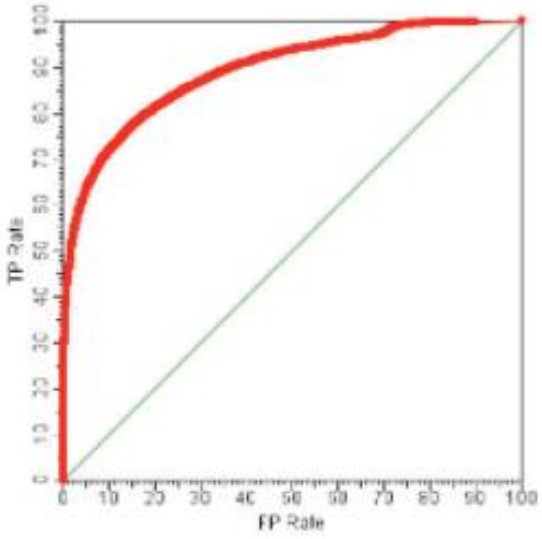


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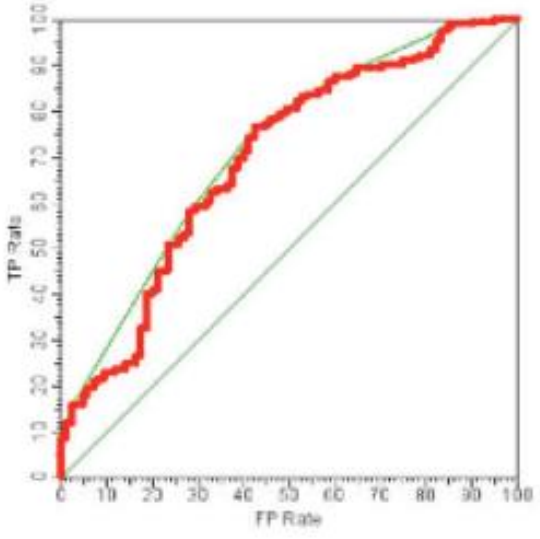
ROC Curves: Examples



Almost perfect classifier



Reasonable classifier



Poor classifier

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ROC Curves: Construction

- Given confidence in positive predictions:
  1. Normalize, so all confidence values are between 0,1.

  2. Enumerate different decision thresholds and find ROC points in the process.

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ROC Curves: Construction

Inputs:	
D	test set
p(i)	ranking of instance i in D, indicates the probability or confidence that the instance i is positive, normalized to [0,1]
P	set of *observed* positive instances in D, where $P \subseteq D$
N	set of *observed* negative instances in D, where $N \subseteq D$

```
for threshold = 1 to 0 by -.01 do
        FP ⇐ 0
        TP ⇐ 0
        for i ∈ D do
                if p(i) ≥ threshold then
                        if i ∈ P then
                                TP ⇐ TP + 1
                        else
                                FP ⇐ FP + 1
                        endif
                endif
        endfor
        tpr ⇐ TP/#P
        fpr ⇐ FP/#N
        Add point (tpr, fpr) to ROC curve
endfor
```


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Case Study: Iris Dataset

(150 flower samples, 3 species, 4 features)



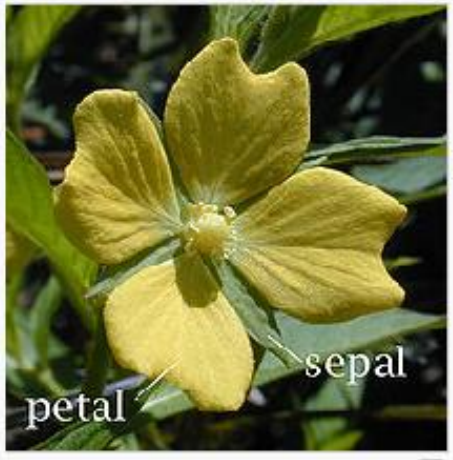
Iris setosa



Iris versicolor

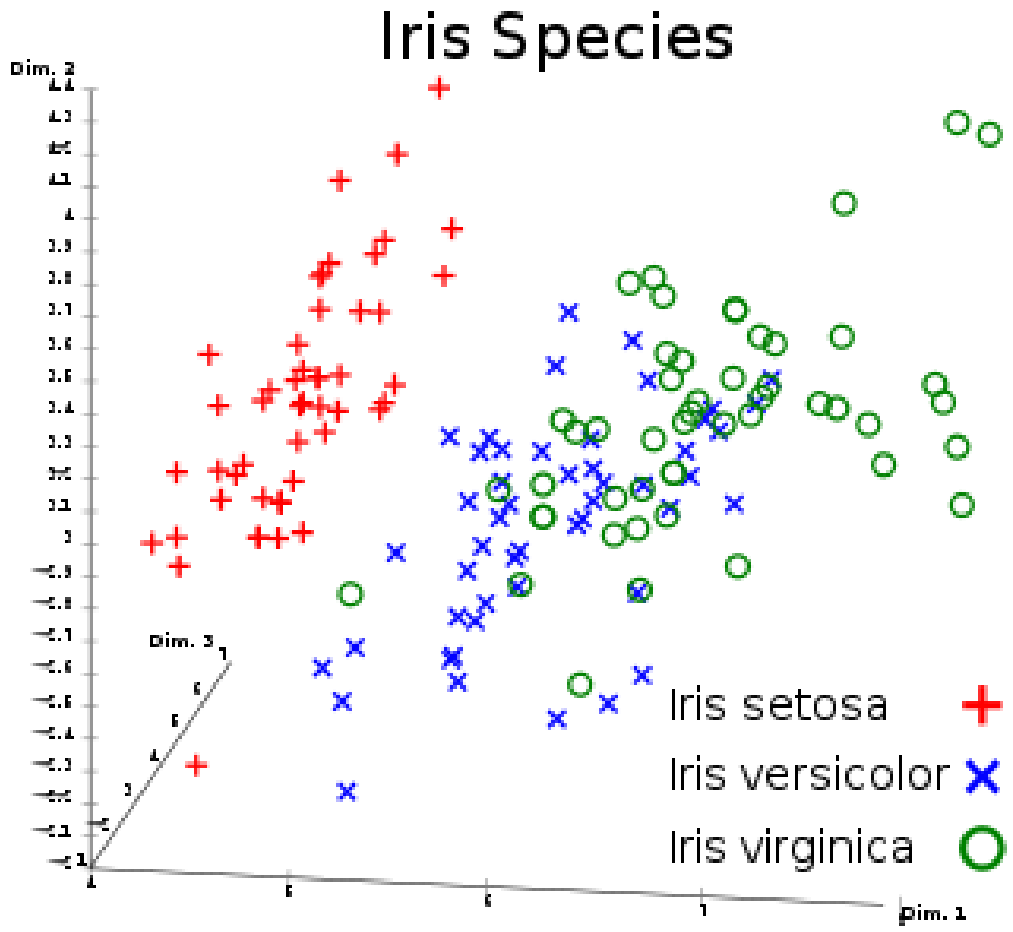


Iris virginica



Features:
Sepal-Petal Length-Width

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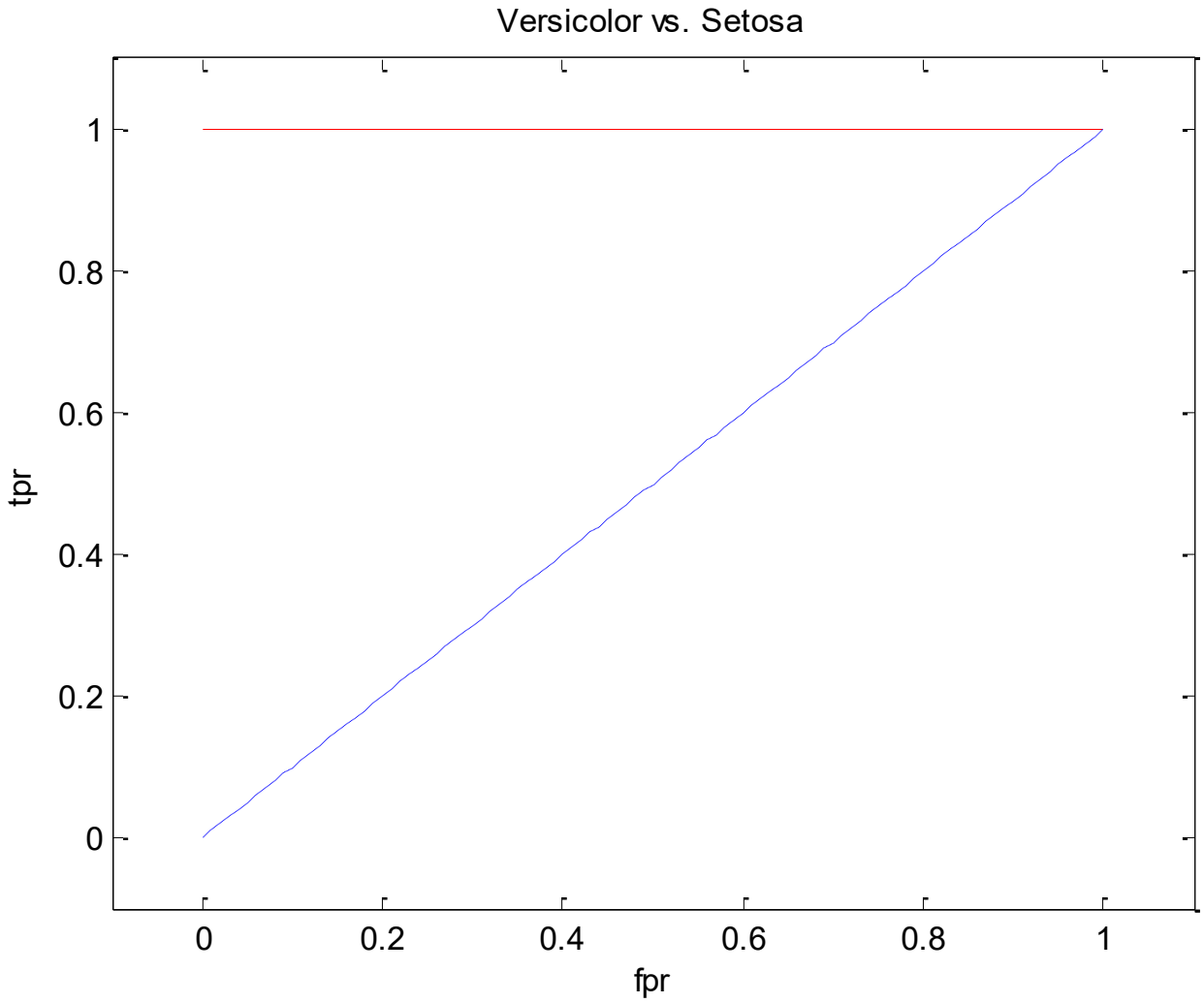
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Iris Dataset: Naïve Bayes classification

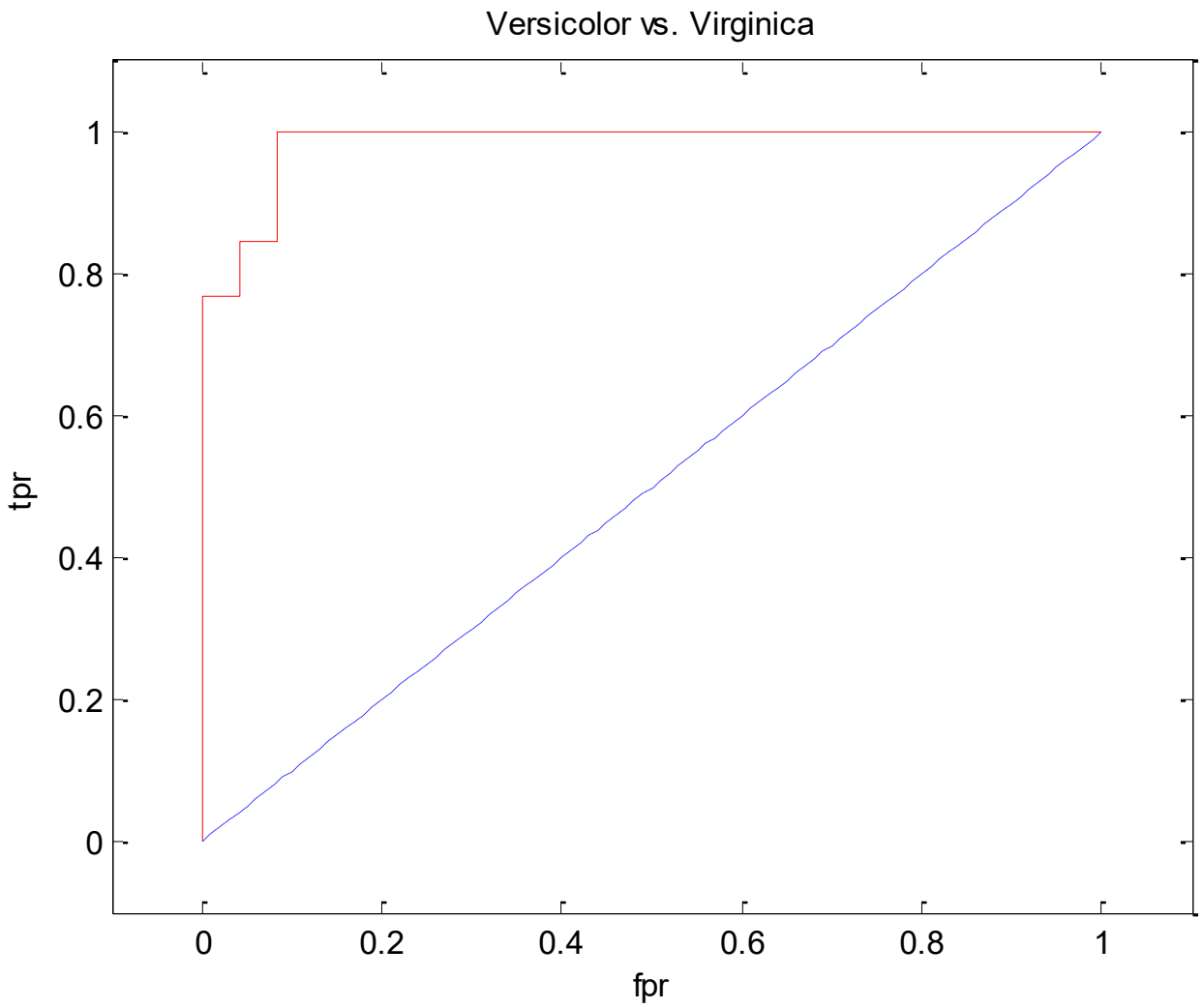
Problem1: Versicolor (+1) vs. Setosa (-1)
Problem2: Versicolor (+1) vs. Virginica (-1)

$$\text{posterior} = \frac{\text{prior x likelihood}}{\text{evidence}}$$

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Thank You!

Questions?